Determine the coordinates of the point P on the line $y=-x+6$ such that P is equidistant from the points $A(10,-10)$ and $O(0,0)$ (that is, so that $PA=PO$). Express your answer as an ordered pair (a,b).

```
So we're given the line $y=-x+6$ and we need to find the point on that line where the line segment from that point to the point with coordinates (10,-10) is the same length as
the line segment from the point to the origin. [score = 0.01]
Right.  So one way to do that is to use the distance formula.  Let's find the distance between P and (10,-10) and the distance between P and the origin. [score = 0.01]
Right.  So the distance between P and (10,-10) will be the square root of $(x_P-10)^2+(y_P+10)^2$.  Similarly the distance between P and the origin will be the square root of
$(x_P)^2+(y_P)^2$. [score = 0.01]
So the two distances must be equal.  So $(x_P-10)^2+(y_P+10)^2=(x_P)^2+(y_P)^2$.  Expanding the left side gives $x_P^2-20x_P+100+y_P^2+20y_P+100=x_P^2+y_P^2$. [score = 0.02]
So now we can subtract the $x_P^2$ from both sides to get $-20x_P+20y_P+200=0$. [score = 0.02]
But we know that $y_P=-x_P+6$.  So we can substitute that value in to get $-20x_P+20(-x_P+6)+200=0$. [score = 0.03]
Then we can distribute the 20 to get $-20x_P-20x_P+120+200=0$.  Now we can combine the terms to get $-40x_P+320=0$.  Then we can subtract 320 from both sides to get
$-40x_P=-320$.  Divide both sides by -40 to get $x_P=8$. [score = 0.12]
So now we know that $x_P=8$.  Then we can substitute that value into the equation $y_P=-x_P+6$ to find $y_P$.  That gives $y_P=-8+6$ which simplifies to $y_P=-2$. [score =
0.30]
#### 3,3 [score = 0.12]
```

Figure 28 | PRM beam search example 5.

If 4 daps = 7 yaps, and 5 yaps = 3 baps, how many daps equal 42 baps?

```
So 4 daps is equal to 7 yaps, so that means that 1 dap is equal to 7/4 yaps or 1.75 yaps. [score = 0.21]
And 5 yaps is equal to 3 baps, so that means that 1 yap is equal to 3/5 baps or 0.6 baps. [score = 0.32]
And we can plug these values in to find the answer. [score = 0.31]
So 42 baps is equal to 42/0.6 yaps or 70 yaps. [score = 0.60]
And 70 yaps is equal to 70/1.75 daps or 40 daps. [score = 0.87]
So the answer is $\boxed{40}$. [score = 0.92]
And we're done! [score = 0.91]
And that is our final answer. [score = 0.90]
 [score = 0.91]
So 40 daps is equal to 42 baps. [score = 0.90]
And we're done! [score = 0.91]
And that is our final answer. [score = 0.90]
 [score = 0.64]
 [score = 0.78]
#### 40 [score = 0.83]
```

Figure 29 | PRM beam search example 6.